

Assignment A1

MATH 147

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1 Assignment A1

1.1 Problem 1

(I) **Prove that $\sqrt{3}$ is irrational.**

We need to prove a lemma first before we prove the proposition.

Lemma 1. *For all $n \in \mathbb{Z}$, if $3 \mid n^2$, then $3 \mid n$.*

Proof. Suppose $3 \nmid n$. There are two cases to discuss.

case 1: $n = 3k + 1$

According to the definition of divisibility, we have

$$n = 3k + 1 \quad \exists k \in \mathbb{Z}.$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3c + 1 \quad c = 3k^2 + 2k \in \mathbb{Z}.$$

thus, $3 \nmid n^2$.

case 2: $n = 3k + 2$

According to the definition of divisibility, we have

$$\exists k \in \mathbb{Z} \text{ s.t. } n = 3k + 2$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3c + 1 \quad c = 3k^2 + 4k + 1 \in \mathbb{Z}.$$

thus, $3 \nmid n^2$. ■

Here is the proof of the question:

Proof. Suppose $\sqrt{3} \in \mathbb{Q}$.

Then, according to the definition of rational numbers,

$$\exists m, n \in \mathbb{Z}, n \neq 0 \quad \text{s.t.} \quad \sqrt{3} = \frac{m}{n}.$$

We assume that m and n have no common factor other than 1. Taking the square of the both sides, we get

$$3 = \frac{m^2}{n^2} \implies m^2 = 3n^2,$$

which implies that $3 \mid m$ according to the lemma 1. According to the definition of divisibility,

$$\exists k \in \mathbb{Z} \quad \text{s.t.} \quad m = 3k.$$

Substituting $m = 3k$ back into $m^2 = 3n^2$, we have

$$\begin{aligned}n^2 &= (3k)^2/3 \\&= 9k^2/3 \\&= 3d, d = k^2 \in \mathbb{Z}.\end{aligned}$$

It means that $3 \mid n$.

Since $3 \mid n$ and $3 \mid m$, it contradicts the assumption that n and m have no common factor other than 1.

Therefore, $\sqrt{3} \in \mathbb{R} \setminus \mathbb{Q}$.

■

(II) *Proof.* We define

$$P := \{x \in \mathbb{R} : x > 0\}$$

to be a nonempty subset of $\mathbb{R}$ that satisfies Trichotomy law, Closure under addition, and closure under multiplication.

Suppose $a, b \in P$. There are two cases to discuss, which are

- $a > b$.
- $a = b$.
- $b < a$.

We will prove case $a > b$ first.

According to the given statement, we have

$$(\forall c > 0) (\exists x > 0) x^2 = c.$$

It follows that there $\exists a', b' \in P$ such that

$$(a')^2 = a \quad \text{and} \quad (b')^2 = b.$$

Since $a > b$, it means that

$$\begin{aligned} a - b &= (a')^2 - (b')^2 \\ &= (a' - b')(a' + b') \in P. \end{aligned}$$

We know that $a' + b' \in P$, so $a' - b' \in P$.

By the ordered field properties, we have

$$(a' - b')^2 \in P \implies (a' - b')^2 > 0.$$

given $a' - b' \in P$ and closure under multiplication.

Expanding the left side, we get

$$(a')^2 - 2a'b' + (b')^2 > 0 \implies \frac{(a')^2 + (b')^2}{2} > a'b'. \quad (1.1)$$

By substituting $(a')^2 = a$ and $(b')^2 = b$ into equation (1.1), we have

$$\frac{a + b}{2} > a'b'.$$

Since $a' \in P$ and $b' \in P$, it follows from the closure under multiplication that

$$a'b' \in P \implies a'b' > 0.$$

Moreover,

$$(a'b')^2 = (a')^2 (b')^2 = ab \in P \implies ab > 0$$

Since $a'b' > 0$ and $ab = a'b'$, therefore, ab is a positive square root of $a'b'$.

$$a'b' = \sqrt{ab}.$$

is the only option.

As a result,

$$\frac{a+b}{2} > \sqrt{ab}.$$

For $b > a$, it can be proved using similar way as $a > b$ by swapping a with b and b with a .

For $a = b$, we have

$$\frac{a+b}{2} = \frac{2a}{2} = a = \sqrt{a^2} = \sqrt{ab}.$$

Therefore, we get

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Thus, the inequality has been proved. ■

Reference: The ordered field properties were consulted in V. A. Zorich, *Mathematical Analysis I*, Section 2.2. Prove style is modified based on the suggestions from my dear father, Changwei Hu.

1.2 Problem 2

1.2.1 (I)

Definition 1. A set $S \subseteq \mathbb{R}$ is called inductive if $1 \in S$ and $k \in S$ implies $k + 1 \in S$.

$\mathbb{R}$ is inductive since $1 \in \mathbb{R}$ and for all $x \in \mathbb{R}$, $x + 1 \in \mathbb{R}$.

$(0, \infty)$ is inductive since $1 \in (0, \infty)$ and for all $x \in (0, \infty)$, $x + 1 \in (0, \infty)$.

$(0, \infty) \setminus \{5\}$ is not inductive since $4 \in (0, \infty) \setminus \{5\}$ but $4 + 1 = 5 \notin (0, \infty) \setminus \{5\}$.

1.2.2 (II)

Proof. Suppose that S and T are inductive sets, and $U = S \cap T$.

By the definition of inductive set, we have

$$1 \in S \quad \text{and} \quad k \in S \implies k + 1 \in S. \quad (1.2)$$

and

$$1 \in T \quad \text{and} \quad k \in T \implies k + 1 \in T. \quad (1.3)$$

Since 1 is in both S and T , it means that $1 \in U$.

By statements (1.2) and (1.3), we get

$$k \in U \iff (k \in S \wedge k \in T) \implies (k + 1 \in S \wedge k + 1 \in T) \iff k + 1 \in U.$$

Thus, U is an inductive set. ■

1.2.3 (III)

Proof. Let $\mathcal{I}$ be the collection of all inductive subsets. We define

$$\mathbb{N} = \bigcap_{S \in \mathcal{I}} S.$$

Since

$$(\forall S \in \mathcal{I}) 1 \in S,$$

it follows that $1 \in \mathbb{N}$.

It suffices to show that

$$k \in \mathbb{N} \implies k + 1 \in \mathbb{N}.$$

Suppose $k \in \mathbb{N}$. By the definition of the intersection of sets, it means that

$$(\forall S \in \mathcal{I}) k \in S.$$

It follows the definition of inductive set that

$$(\forall S \in \mathcal{I}) k + 1 \in S,$$

which implies

$$k + 1 \in \mathbb{N}.$$

Thus, $\mathbb{N}$ is inductive. ■

1.3 Problem 3

1.3.1 (I)

Proof. Suppose that $P(1)$ is true and

$$(\forall n \in \mathbb{N}) [(P(1) \wedge P(2) \wedge \cdots \wedge P(n)) \implies P(n+1)].$$

Let

$$T(n) := P(1) \wedge P(2) \wedge \cdots \wedge P(n).$$

Since $P(1)$ is true, it implies that $T(1)$ is also true. We also assume $T(n)$ is true.

By the assumption, we obtain

$$T(n) \implies P(n+1),$$

and hence, since $T(n)$ is already assumed to be true,

$$T(n) \wedge P(n+1)$$

is also true.

By the definition of $T(n+1)$, it follows that

$$T(n) \implies T(n+1).$$

Following the Principle of Mathematical Induction, this yields

$$(\forall n \in \mathbb{N}) T(n) \implies (\forall n \in \mathbb{N}) P(n).$$

Thus, Principle of Strong Mathematical Induction is correct.

■

1.3.2 (II)

Proof. We define S is a set such that

$$S := \{n \in \mathbb{N} \cup \{0\} : p_n(x) \text{ is a degree } n \text{ polynomial and the coefficient of } x^n \text{ in } p_n(x) \text{ is } 1\}$$

We know $0 \in S$ is true since $p_0(x) = 1x^0 = 1 \cdot 1 = 1$ is a degree 0 polynomial and the coefficient of x^0 is 1.

We also know that $1 \in S$ is true since $p_1(x) = 1 \cdot x$ is a degree 1 polynomial and the coefficient of x is 1.

It is suffice to show that

$$(n - 1 \in S) \wedge (n \in S) \implies (n + 1 \in S).$$

Suppose $n - 1 \in S$ and $n \in S$ are true. We define

$$p_{n-1}(x) = x^{n-1} + q(x) \quad \text{and} \quad p_n = x^n + r(x),$$

where

$$\deg r \leq n - 1 \quad \text{and} \quad \deg q \leq n - 2.$$

Then, by the given statement

$$\begin{aligned} p_{n+1}(x) &= x \cdot p_n(x) - p_{n-1}(x) \\ &= x \cdot (x^n + r(x)) - (x^{n-1} + q(x)) \\ &= x^{n+1} + xr(x) - x^{n-1} - q(x). \end{aligned}$$

Since the term $xr(x) - x^{n-1} - q(x)$ has at most degree n , the term x^{n+1} cannot be canceled. Therefore, $p_{n+1}(x)$ is a degree $n + 1$ polynomial and the coefficient of x^{n+1} is 1

Thus, by the Principle of Complete Mathematical Induction,

$$\forall n \in \mathbb{N} \cup \{0\}, n \in S,$$

which implies that the given statement is true. ■

1.3.3 III

Proof. Let $0 < |t| < 1$ and define statement

$$T(n) : p_n(t + t^{-1}) = \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}}.$$

We know that $T(0)$ is true since

$$p_0(t + t^{-1}) = \frac{t^{0+1} - t^{0-1}}{t - t^{-1}} = \frac{t - t^{-1}}{t - t^{-1}} = 1.$$

We also know that $T(1)$ is true since

$$p_1(t + t^{-1}) = \frac{t^{1+1} - t^{-1-1}}{t - t^{-1}} = \frac{t^2 - t^{-2}}{t - t^{-1}} = \frac{(t^1 - t^{-1})(t^1 + t^{-1})}{t - t^{-1}} = t + t^{-1}.$$

Suppose that $T(n-1)$ and $T(n)$ are true. It suffices to show that

$$(T(n-1) \wedge T(n)) \implies T(n+1).$$

Following from the given statement,

$$\begin{aligned} p_{n+1}(t + t^{-1}) &= (t + t^{-1}) p_n(t + t^{-1}) - p_{n-1}(t + t^{-1}) \\ &= (t + t^{-1}) \cdot \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} - \frac{t^{n-1+1} - t^{-(n-1)-1}}{t - t^{-1}} \\ &= (t + t^{-1}) \cdot \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} - \frac{t^n - t^{-n}}{t - t^{-1}} \\ &= \frac{t^{n+2} - t^{-n+2}}{t - t^{-1}} \\ &= \frac{t^{(n+1)+1} - t^{-(n+1)-1}}{t - t^{-1}} \end{aligned}$$

This implies $T(n+1)$ is true.

By the Principle of Complete Mathematical Induction, we get

$$(\forall n \in \mathbb{N} \cup \{0\}) T(n).$$

Thus,

$$(\forall n \in \mathbb{N} \cup \{0\}) \left[p_n(t + t^{-1}) = \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} \right].$$

■

1.3.4 IV

Proof. Given function

$$g : D \rightarrow \mathbb{R}, \quad g(t) := t + t^{-1}$$

where

$$D := \{x \in \mathbb{R} : 0 < |x| < 1\}.$$

Let y be arbitrary in $\text{rng}(g)$, then

$$(\exists t \in D) y = t + t^{-1}.$$

Since $t \neq 0$, we get

$$y = t + \frac{1}{t} \iff ty = t^2 + 1 \iff 0 = t^2 - yt + 1.$$

By the definition of the set D , element t must be in real number set. So the equation

$$0 = t^2 - yt + 1$$

must have real roots. Therefore,

$$(-y)^2 - 4 \cdot (1) \cdot (1) \geq 0 \implies y^2 - 4 \geq 0 \implies y^2 \geq 4 \implies |y| \geq 2.$$

However, $|y| \neq 2$ since

$$t = \frac{-(-y) \pm \sqrt{(-y)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-(2) \pm \sqrt{(-2)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-2}{2} = -1$$

and

$$t = \frac{-(-y) \pm \sqrt{(-y)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-(-2) \pm \sqrt{(2)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{2}{2} = 1$$

which makes $t \notin D$. Thus, $|y| > 2$, which also implies $\text{rng}(g) = (-\infty, -2) \cup (2, \infty)$.

We want to show that

$$(\forall t \in (-1, 0) \cup (0, 1)) p_n(t + t^{-1}) \neq 0$$

Suppose, for contradiction, $p_n(t + t^{-1}) = 0$. We have

$$\begin{aligned} \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} &= 0 \\ \Rightarrow t^{n+1} - t^{-n-1} &= 0 \\ \Rightarrow t^{n+1} &= t^{-n-1} \\ \Rightarrow t^{2n+2} &= t^0 = 1 \end{aligned}$$

Since $2n + 2 > 0$ and $0 < |t| < 1$,

$$0 < |t|^{2n+2} < 1.$$

This is a contradiction since t will never be 1.

Thus, for all $t \in (-1, 0) \cup (0, 1)$, $p_n(t + t^{-1}) \neq 0$.

Since $x = g(t) = t + t^{-1}$, this yields

$$(\forall x \in \text{rng}(g))p_n(x) \neq 0 \quad \implies \quad ((\forall x \in \mathbb{R} \setminus [-2, 2])p_n(x) \neq 0)$$

It follows that

$$(\forall x \in [-2, 2])p_n(x) = 0$$

Thus, the claim has been proved. ■

1.4 Problem 4

1.4.1 I

Proof. Suppose

$$(\forall x > 0) (\exists n \in \mathbb{N}) \frac{1}{n} < x.$$

Multiplying both sides by $\frac{n}{x}$,

$$\frac{1}{x} < n.$$

Since every positive real number has a positive multiplicative inverse, we have

$$(\forall x > 0) (\exists x' > 0) x \cdot x' = 1. \quad \implies \quad x' = \frac{1}{x} > 0$$

This yields

$$(\forall x > 0) (\exists x' > 0) (\exists n \in \mathbb{N}) \frac{1}{x} = x' < n,$$

Now, Since we also have

$$(\forall x' > 0) (\exists x > 0) x' = \frac{1}{x} > 0,$$

this yields

$$(\forall x' > 0) (\exists n \in \mathbb{N}) x' < n.$$

Thus, the Archimedean property is proved. ■

1.4.2 ii

Proof for the case for $x \in \mathbb{Q}$

We need to prove

$$(\forall x \in \mathbb{Q}) \left[(x > 0) \implies (\exists n \in \mathbb{N}) \frac{1}{n} < x \right].$$

Proof. Since $x \in \mathbb{Q}$, by the definition

$$(\forall x \in \mathbb{Q}) (\exists m, d \in \mathbb{Z}) \left[d \neq 0 \implies x = \frac{m}{d} \right].$$

Since $x > 0$, which implies $m \neq 0$.

Consider the case where both m and d are positive first. We have

$$\frac{1}{n} < \frac{m}{d} \implies d < mn.$$

Since $d \geq 1$ and $m \geq 1$, we can set

$$n = d + 1.$$

This yields,

$$d \leq md < m(d + 1) = md + m.$$

Therefore,

$$d < mn \implies \frac{1}{n} < \frac{m}{d}.$$

Thus, the original claim holds.

Consider the case where $m, d < 0$.

Since

$$\frac{m}{d} = \frac{-m}{-d},$$

we are able to make

$$m' = -m \quad \text{and} \quad d' = -d$$

Then treat them as the case where m and d are both positive. ■

Prove for the case where $x > 0, x = \sqrt{y}, \exists y \in \mathbb{Q}$. We need to prove

$$(\forall x > 0) \left[[(\exists y \in \mathbb{Q}) [y > 0 \wedge x = \sqrt{y}]] \implies (\exists n \in \mathbb{N}) \frac{1}{n} < x \right].$$

Proof. Let

$$y = \frac{m}{d}, \exists m \in \mathbb{Z}, \exists d \in \mathbb{Z}, d \neq 0.$$

Since $y > 0$, which implies

$$m \neq 0,$$

and we have two cases to discuss. They are

$$(m \geq 1) \wedge (d \geq 1) \quad \text{and} \quad (m \leq -1) \wedge (d \leq -1).$$

- Case $(m \geq 1) \wedge (d \geq 1)$:

$$\begin{aligned} (\exists n \in \mathbb{N}) \frac{1}{n} < x &\Leftrightarrow (\exists n \in \mathbb{N}) \frac{1}{n} < \sqrt{y} = \sqrt{\frac{m}{d}} \\ &\Leftrightarrow (n \in \mathbb{N}) \frac{1}{n^2} < \frac{m}{d} \\ &\Leftrightarrow (\exists n \in \mathbb{N}) d < mn^2, \end{aligned}$$

Since $m \geq 1$, we can select $n = (d + 1)$. We have

$$mn^2 = m(d + 1)^2 = m(d^2 + 2d + 1) = md^2 + 2md + m > d$$

Thus, the claim hold.

- Case $(m \leq -1) \wedge (d \leq -1)$

Since $y > 0$, we are able to multiple y by $\frac{-1}{-1} = 1$, where we have

$$y = \frac{m}{d} \cdot 1 = \frac{m}{d} \cdot \frac{-1}{-1} = \frac{-m}{-d}, -m \geq 1, -d \geq 1.$$

Then, we treat this case similar to the last case.

Thus, the claim holds. ■

1.4.3 III

Proof. Suppose x is an arbitrary positive number in $\mathbb{R}$. We define the set

$$T := \{m \in \mathbb{N} : x < m\}.$$

By Archimedean Property, we know that $T \neq \emptyset$. By well ordering principle, since $T \neq \emptyset$ and $T \subset \mathbb{N}$, T has a least element. We define

$$k := \min T.$$

Since k is the least element in T , which means

$$k > x \quad \text{and} \quad k - 1 \notin T.$$

The definition of set T implies

$$k - 1 \leq x.$$

Thus, we have

$$k - 1 \leq x < k.$$

Let $n = k - 1$, we replace k by n .

$$n \leq x < n + 1.$$

Thus, the original claim holds.

For $x < 0$ in $\mathbb{R}$. You are able to add a natural number $t > -x$ which makes $t + x$ positive. Following the similar steps as $x > 0 \in \mathbb{R}$. We prove there exists a number n in $\mathbb{N}$ such that

$$n \leq t + x < n + 1.$$

Subtracting t , we get

$$n - t \leq x < n - t + 1.$$

Let us define $z = n - t$, we get

$$z \leq x < z + 1$$

where $z \in \mathbb{Z}$.

■

At here, we prove the uniqueness of such integer m .

Proof. Suppose there exists two integers n and m in $\mathbb{N}$ such that

$$n \leq x < n + 1 \quad m \leq x < m + 1$$

and

$$n \neq m.$$

Without loss of generality, we assume $n < m$. We have

$$n + 1 \leq m.$$

Then, we get

$$n \leq x < n + 1 \leq m \leq x < m + 1.$$

This yields,

$$x < x,$$

which contradicts the fact that $x = x$.

Thus, $n = m$. Therefore, the integer m is unique. ■

1.4.4 IV

Proof. Suppose x and y are both arbitrary number in $\mathbb{R}$ with $y > x \geq 0$.

By hypothesis,

$$y - x > 0.$$

Following Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$y - x > \frac{1}{n} > 0.$$

This yields

$$ny - nx > 1,$$

which implies

$$ny > 1 + nx.$$

Moreover, by Archimedean Property, there exists $m \in \mathbb{N}$ such that

$$m - 1 \leq nx < m.$$

Adding 1, we obtain

$$m \leq nx + 1 < m + 1.$$

Hence

$$m - 1 \leq nx < m \leq nx + 1 < ny,$$

which implies

$$nx < m < ny.$$

Dividing n ,

$$x < \frac{m}{n} < y.$$

By definition,

$$x < q < y$$

where $q = \frac{m}{n} \in \mathbb{Q}$.

If

$$x < 0,$$

then there exists a natural number k with $k > -x$ such that

$$(\exists m, n \in \mathbb{N}) x + k < \frac{m}{n} < y + k,$$

where $x + k > 0$ and $y + k > 0$.

Subtracting k

$$x < \frac{m}{n} - k < y.$$

We have

$$x < \frac{m - nk}{n} < y$$

By definition, there exists q such that

$$x < \frac{m - nk}{n} = q < y$$

where $m - nk \in \mathbb{Z}$ and $n \in \mathbb{N}$.

This proves the claim. ■

1.5 Problem 5

1.5.1 (i)

Proof. Suppose

$$S \subset \mathbb{R}, S \neq \emptyset \quad \text{and} \quad T \subset \mathbb{R}, T \neq \emptyset,$$

and suppose S and T are bounded above.

We define

$$S + T := \{s + t : s \in S \text{ and } t \in T\}.$$

It suffices to show that

$$\sup(S + T) \leq \sup S + \sup T,$$

and

$$\sup(S + T) \geq \sup S + \sup T.$$

We start with the first one.

By the definition, we have

$$(\forall s \in S) \sup S \geq s \quad \text{and} \quad (\forall t \in T) \sup T \geq t.$$

Adding them together

$$\sup S + \sup T \geq s + t$$

for every $s \in S$ and $t \in T$.

Hence, $\sup S + \sup T$ is an upper bound of $S + T$. Since $\sup(S + T)$ is the least upper bound of $S + T$, we have

$$\sup(S + T) \leq \sup S + \sup T.$$

Next, let us prove the second one.

Let $t \in T$ be arbitrary. For every $s \in S$, we have

$$s + t \in S + T.$$

We already know

$$\sup(S + T) \geq s + t.$$

This yields

$$\sup(S + T) - t \geq s$$

for every $s \in S$.

Since $\sup(S + T) - t$ is an upper bound of S ,

$$\sup(S + T) - t \geq \sup S.$$

Moreover,

$$\sup(S + T) - \sup S \geq t.$$

Since $t \in T$ was arbitrary, $\sup(S + T) - \sup S$ is an upper bound of T . Hence,

$$\sup(S + T) - \sup S \geq \sup T.$$

Hence, we proved

$$\sup(S + T) \geq \sup S + \sup T.$$

Thus, the claim holds. ■

1.5.2 II

Proof. Suppose

$$S := \left\{ \frac{1}{p} : p \text{ is a prime number} \right\}.$$

We claim $\sup S = \frac{1}{2}$ and $\inf S = 0$.

In order to show $\inf S = 0$, it suffices to prove that

$$\inf S \leq 0 \quad \text{and} \quad \inf S \geq 0.$$

We start with the first one. Suppose, for contradiction, $w = \inf S > 0$. By Archimedean Property,

$$(\exists n \in \mathbb{N}) 0 < \frac{1}{n} < w.$$

Since for each $n \in \mathbb{N}$, there is a prime number p such that $p \geq n$. We get

$$(\forall n \in \mathbb{N}) (\exists p \in \text{prime number set}) p \geq n.$$

Hence,

$$(\exists n \in \mathbb{N}) (\exists p \in \text{prime number set}) 0 < \frac{1}{p} \leq \frac{1}{n} < w.$$

By hypothesis

$$(\exists s \in S) 0 < \frac{1}{p} < w.$$

This contradicts that w is a greatest lower bound, since w is not even a lower bound.

Let us take a look at the second one.

Suppose, for contradiction, $w = \inf S < 0$.

Since 0 is already a lower bound of S by the Archimedean Property, w must not be a greatest lower bound.

The claim holds.

In order to show that $\sup S = \frac{1}{2}$, we want to show that

$$\sup S \geq \frac{1}{2} \quad \text{and} \quad \sup S \leq \frac{1}{2}.$$

Suppose, for contradiction, $w = \sup S < \frac{1}{2}$.

There exists an ϵ such that

$$w < w + \epsilon.$$

We pick the $\epsilon = \frac{1}{2} - w$.

$$w < w + \epsilon = w + \frac{1}{2} - w = \frac{1}{2}.$$

Since 2 is prime, so $\frac{1}{2}$ is in S . Thus, w is not an upper bound. This contradicts to the fact that w is a least upper bound.

Suppose, for contradiction, $w = \sup S > \frac{1}{2}$.

Since 2 is the least prime number, for all prime number p ,

$$p \geq 2.$$

Hence,

$$\frac{1}{p} \leq \frac{1}{2}.$$

Therefore, $\frac{1}{2}$ is an upper bound.

It contradicts to our assumption that $w > \frac{1}{2}$ is a least upper bound.

Therefore, the claim holds. ■

1.6 P6

1.6.1 I

Proof. let $\epsilon > 0$ be arbitrary in $\mathbb{R}$, we want to find n_0 in $\mathbb{N}$ such that for all $n \geq n_0$ in $\mathbb{N}$,

$$|x_n - 4| < \epsilon.$$

By Archidemean Property, there exists a n in $\mathbb{N}$ such that

$$\frac{1}{n} < \frac{\epsilon}{60}.$$

This equals to

$$\frac{60}{n} < \epsilon.$$

Indeed, when $n \geq 2$, $n^3 - 6 \geq \frac{1}{4}n^3$.

We have

$$\begin{aligned} \left| \frac{4n^3 + 3n}{n^3 - 6} - 4 \right| &= \left| \frac{3n + 24}{n^3 - 6} \right| \leq \left| \frac{3n + 12n}{n^3 - 6} \right| = \frac{3n + 12n}{n^3 - 6} \\ &\leq \frac{15n}{\frac{1}{4}n^3} = \frac{60n}{n^3} = \frac{60}{n^2} \leq \frac{60}{n} < \epsilon. \end{aligned}$$

when $n \geq \max(n_0, 2)$.

Therefore, we prove

$$(\forall \epsilon \in \mathbb{R}) (\exists n \in \mathbb{N}) \left[(\forall n \geq n_0) \implies \left(\left| \frac{4n^3 + 3n}{n^3 - 6} - 4 \right| < \epsilon \right) \right].$$

■

Remark. The idea of the solution comes from "Elementary Analysis" by Kenneth A. Ross.

1.6.2 II

Proof. Suppose $\{a_n\}$ be a convergent sequence of integers.

If $a_n \neq a_{n_0}$,

$$|a_n - a_{n_0}| \geq 1.$$

By hypothesis, since $\{a_n\}$ is a convergent sequence,

$$(\forall \epsilon > 0) (\exists n_0 \in \mathbb{N}) [(\forall n \geq n_0) \implies (|a_n - L| < \epsilon)]$$

We pick $\epsilon = \frac{1}{2}$. By definition, there exists N in $\mathbb{N}$ such that

$$(\forall n \geq N) |x_n - L| < \frac{1}{2}.$$

Suppose $n, m \geq N$, we have

$$|a_n - a_m| \leq |a_n - L| + |a_m - L| < \frac{1}{2} + \frac{1}{2} = 1.$$

Since when $a_n \neq a_m$, their difference must be greater or equal to 1.

Hence, $|a_n - a_m| = 0$.

Thus, there exists a n_0 in $\mathbb{N}$ such that for all $n \geq N$, $a_n = a_{n_0}$. ■